

D Primal Integral Results with Overhead

When integrating the **BackPaS** pipeline overhead (Graph construction, GNN prediction and Trust Region construction times) the $PI(1000)$ results vary slightly. Table 9 presents the results including these overhead for all **BackPaS** variants. In all datasets (MIS, CA, MVC) **BackPaS** remain better than **Gurobi** and **Conpas**. More differences can be seen between the **BackPaS** variants, where **BackPaS-Net** had a greater penalization due to the longer time required for the literal-based bipartite graph. Although we believe the overheads could be reduced by optimizing the code.

Table 9. Anytime Performance (Primal Integral) Comparison on Test Benchmarks. including BACKPAS pipeline overhead Mean Primal Integral (PI) values, with standard deviations (in parentheses), are reported for the CA, MIS, and MVC test partitions. *Lower PI values indicate better anytime performance.* The ranks provided (e.g., 1st) are determined by pairwise Wilcoxon signed-rank tests ($p < 0.05$) corrected for multiple tests using Benjamini–Hochberg correction. Methods sharing the same rank are not statistically significantly different. A method at rank n performs statistically better than all methods at ranks $> n$, and statistically worse than all methods at ranks $< n$.

Method	CA	MIS	MVC
Gurobi	20.84 (6.15) 2 nd	44.65 (7.81) 5 th	7.79 (1.15) 6 th
ConPaS	19.67 (7.45) 2 nd	9.10 (1.98) 4 th	6.84 (2.53) 5 th
BackPaS	16.66 (6.84) 1 st	2.54 (0.56) 1 st	2.48 (0.66) 2 nd
BackPaS-Param	30.86 (9.79) 4 th	5.30 (1.96) 3 rd	2.28 (1.12) 1 st
BackPaS-Net	24.73 (10.58) 3 rd	3.09 (1.94) 2 nd	3.17 (0.61) 4 th
BackPaS-V0	17.35 (6.82) 1 st	2.99 (0.64) 2 nd	2.83 (0.44) 3 rd